# onlineretail2_1

```
%pyspark
df_zeppelin = spark.read.parquet("/onlineretail2/onlineretail_ii_v1.parquet")
```

READY

```
%pyspark
df_zeppelin.show(5)
```

READY

```
+-------+---------+--------------------+--------+-------------------+-----+----------+--------------+------------------+------------------+
|Invoice|StockCode|         Description|Quantity|        InvoiceDate|Price|Customer_ID|       Country|       Total_Sales|   Log_Total_Sales|
+-------+---------+--------------------+--------+-------------------+-----+----------+--------------+------------------+------------------+
| 489434|    85048|15CM CHRISTMAS GL...|      12|2009-12-01 07:45:00| 6.95|   13085.0|United Kingdom|              83.4|1.9263424466256551|
| 489434|   79323P|  PINK CHERRY LIGHTS|      12|2009-12-01 07:45:00| 6.75|   13085.0|United Kingdom|              81.0|1.9138138523837167|
| 489434|   79323W| WHITE CHERRY LIGHTS|      12|2009-12-01 07:45:00| 6.75|   13085.0|United Kingdom|              81.0|1.9138138523837167|
| 489434|    22041|RECORD FRAME 7" S...|      48|2009-12-01 07:45:00|  2.1|   13085.0|United Kingdom|100.80000000000001| 2.00774777800074|
| 489434|    21232|STRAWBERRY CERAMI...|      24|2009-12-01 07:45:00| 1.25|   13085.0|United Kingdom|              30.0|1.4913616938342726|
+-------+---------+--------------------+--------+-------------------+-----+----------+--------------+------------------+------------------+
only showing top 5 rows
```

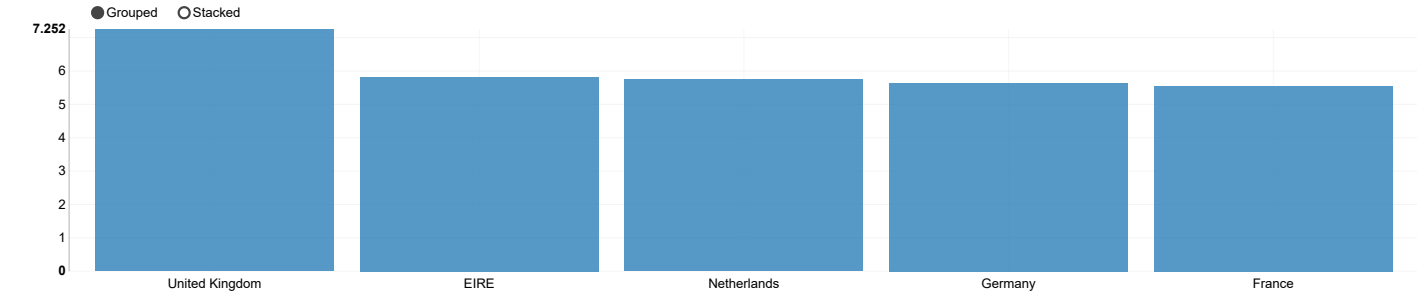
```
%pyspark
#convert to SQL Table view
df_zeppelin.createOrReplaceTempView("online_retail")
```

READY

```
%sql
--TESTING
SELECT Country, LOG10(SUM(Total_Sales)) as Log_Total_Sales
FROM online_retail
GROUP BY Country
ORDER BY Log_Total_Sales DESC
LIMIT 10
```

READY

settings




```
%sql
--Check Missing Values
SELECT
    COUNT(*) - COUNT(Invoice) AS Invoice_null,
    COUNT(*) - COUNT(StockCode) AS StockCode_null,
    COUNT(*) - COUNT(Description) AS Description_null,
    COUNT(*) - COUNT(Quantity) AS Quantity_null,
    COUNT(*) - COUNT(InvoiceDate) AS InvoiceDate_null,
    COUNT(*) - COUNT(Price) AS Price_null,
    COUNT(*) - COUNT(Customer_ID) AS CustomerID_null,
    COUNT(*) - COUNT(Country) AS Country_null
FROM online_retail
```

READY

| Invoice_null | StockCode_null | Description_null | Quantity_null | Inv |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |

```
%sql
--create lookup table to map the description
CREATE OR REPLACE TEMPORARY VIEW stock_desc AS
SELECT
    StockCode,
    MAX(Description) AS Description
FROM online_retail
WHERE Description IS NOT NULL AND Description != ''
GROUP BY StockCode
```

READY

```
%sql

--impute the missing value
CREATE OR REPLACE TEMPORARY VIEW online_retail_cleaned AS
SELECT
    main.Invoice,
    main.StockCode,
    COALESCE(
        NULLIF(main.Description, ''),
        lookup.Description,
        'NA'
    ) AS Description,
    main.Quantity,
    main.InvoiceDate,
    main.Price,
    main.Customer_ID,
    main.Country
FROM online_retail main
LEFT JOIN stock_desc lookup
  ON main.StockCode = lookup.StockCode
```

READY

```
%sql

--review
SELECT COUNT(*) AS remaining_missing_descriptions
FROM online_retail_cleaned
WHERE Description IS NULL OR Description = ''
```

READY

settings

**No Data A**

```
%sql

-- Check if description ada unique ke
SELECT
    StockCode,
    COUNT(DISTINCT Description) AS Total_Unique_Descriptions,
    COLLECT_SET(Description) AS List_Of_Descriptions
FROM online_retail
GROUP BY StockCode
HAVING COUNT(DISTINCT Description) > 1
```

READY

| StockCode | Total_Unique_Descriptions |
|---|---|

```
%sql

-- clean dup product name


CREATE OR REPLACE TEMPORARY VIEW stock_desc_lookup AS
SELECT
    StockCode,
    MAX(Description) AS Master_Description
FROM online_retail
WHERE Description IS NOT NULL
  AND Description NOT IN ('NA', '', 'nan', '?') -- tapis sisa awal
GROUP BY StockCode
```

READY

```
%sql

-- Check if description ada unique ke
SELECT
    StockCode,
    COUNT(DISTINCT Description) AS Total_Unique_Descriptions,
    COLLECT_SET(Description) AS List_Of_Descriptions
FROM online_retail
GROUP BY StockCode
HAVING COUNT(DISTINCT Description) > 1
```

READY

| StockCode | Total_Unique_Descriptions | List_Of_Descriptions |
|---|---|---|
| 21249 | 2 | WrappedArray(WOOD ANIMALS HEIGHT CHART STICKERS, WOODLAND HEIGHT CHART STICKERS) |
| 21535 | 2 | WrappedArray(RETRO SPOT SMALL MILK JUG, RED RETROSPOT SMALL MILK JUG) |
| 21711 | 2 | WrappedArray(FOLDING UMBRELLA WHITE/RED POLKADOT, FOLDING UMBRELLA WHITE/RED SPOT) |
| 23236 | 4 | WrappedArray(DOILEY BISCUIT TIN, DOILEY STORAGE TIN, STORAGE TIN VINTAGE DOILEY, STORAGE TIN VINTAGE |
| 22529 | 2 | WrappedArray(MAGIC DRAWING SLATE GO TO THE FAIR, MAGIC SLATE GO TO THE FAIR) |
| 82486 | 2 | WrappedArray(WOOD S/3 CABINET ANT WHITE FINISH, 3 DRAWER ANTIQUE WHITE WOOD CABINET) |
| 23462 | 2 | WrappedArray(ROCOCO WALL MIROR, ROCOCO WALL MIRROR WHITE) |
| 23535 | 3 | WrappedArray(WALL ART BICYCLE SAFETY, WALL ART BICYCLE SAFTEY, BICYCLE SAFTEY WALL ART) |
| 48184 | 2 | WrappedArray(DOORMAT ENGLISH ROSE, DOOR MAT ENGLISH ROSE) |

```
%sql
-- 1. Redo Bina Lookup Table becauase output above still show duplicate product name
CREATE OR REPLACE TEMPORARY VIEW stock_desc_lookup AS
SELECT
    StockCode,
    MAX(Description) AS Master_Description
FROM online_retail
WHERE Description IS NOT NULL
  AND Description NOT IN ('NA', '', 'nan', '?', '????damages????', '?display?', 'display')
GROUP BY StockCode
```

READY

READY

```
%sql
-- 2. Map nama produk dan tapis gross sales
CREATE OR REPLACE TEMPORARY VIEW online_retail_clean AS
SELECT
    main.Invoice,
    main.StockCode,
    COALESCE(lookup.Master_Description, main.Description, 'NA') AS Description,
    main.Quantity,
    main.InvoiceDate,
    main.Price,
    main.Customer_ID,
    main.Country,
    main.Total_Sales,
    main.Log_Total_Sales
FROM online_retail main
LEFT JOIN stock_desc_lookup lookup
  ON main.StockCode = lookup.StockCode
WHERE main.Total_Sales > 0
```

READY

```
%sql

-- Check if description ada unique ke
SELECT
    StockCode,
    COUNT(DISTINCT Description) AS Total_Unique_Descriptions,
    COLLECT_SET(Description) AS List_Of_Descriptions
FROM online_retail
GROUP BY StockCode
HAVING COUNT(DISTINCT Description) > 1
```

READY

| StockCode | Total_Unique_Descriptions |
|---|---|

```
%sql
-- check rows with duplicates
SELECT
    Invoice,
    StockCode,
    Description,
    Quantity,
    InvoiceDate,
    Price,
    Customer_ID,
    Country,
    COUNT(*) as row_count
FROM online_retail_cleaned
GROUP BY
    Invoice,
    StockCode,
    Description,
    Quantity,
    InvoiceDate,
    Price,
    Customer_ID,
    Country
HAVING COUNT(*) > 1
```

READY

| Invoice | StockCode | Description | Quantity | I |
|---|---|---|---|---|
| 490300 | 21738 | COSY SLIPPER SHOES SMALL RED | 1 | 2 |
| 491204 | 47591B | SCOTTIES DES CHILD'S APRON | 2 | 2 |
| 496651 | 47568 | ENGLISH ROSE DESIGN PEG BAG | 1 | 2 |
| 498567 | 21976 | PACK OF 60 MUSHROOM CAKE CASES | 1 | 2 |
| 499973 | 85123A | WHITE HANGING HEART T-LIGHT HOLDER | 2 | 2 |
| 501900 | 21897 | POTTING SHED CANDLE CITRONELLA | 1 | 2 |
| 502189 | 21559 | STRAWBERRY LUNCHBOX WITH CUTLERY | 1 | 2 |
| 502660 | 17021 | NAMASTE SWAGAT INCENSE | 18 | 2 |
| 505177 | 22410 | TIN CAN HOUSEKEEPING DESIGN | 2 | 2 |

**Output is truncated** to 1000 rows. Learn more about **zeppelin.spark.maxResult**

READY

```
%sql
-- sum the total number of duplicate rows
WITH duplicated_groups AS (
    SELECT COUNT(*) as row_count
    FROM online_retail_cleaned
    GROUP BY
        Invoice,
        StockCode,
        Description,
        Quantity,
        InvoiceDate,
        Price,
        Customer_ID,
        Country
    HAVING COUNT(*) > 1
)
SELECT SUM(row_count) AS total_duplicate_rows
FROM duplicated_groups
```

| total_duplicate_rows |
|---|
| 66098 |

READY

```
%sql
-- Drop duplicates & simpan ke dalam view baru yang bersih
CREATE OR REPLACE TEMPORARY VIEW online_retail_final AS
SELECT DISTINCT
    Invoice,
    StockCode,
    Description,
    Quantity,
    InvoiceDate,
    Price,
    Customer_ID,
    Country
FROM online_retail_cleaned
```

READY

```
%sql
-- recheck duplicate
WITH duplicated_groups AS (
    SELECT COUNT(*) as row_count
    FROM online_retail_final
    GROUP BY
        Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer_ID, Country
    HAVING COUNT(*) > 1
)
SELECT COALESCE(SUM(row_count), 0) AS total_duplicate_rows
FROM duplicated_groups
```

| total_duplicate_rows |
|---|
| 0 |

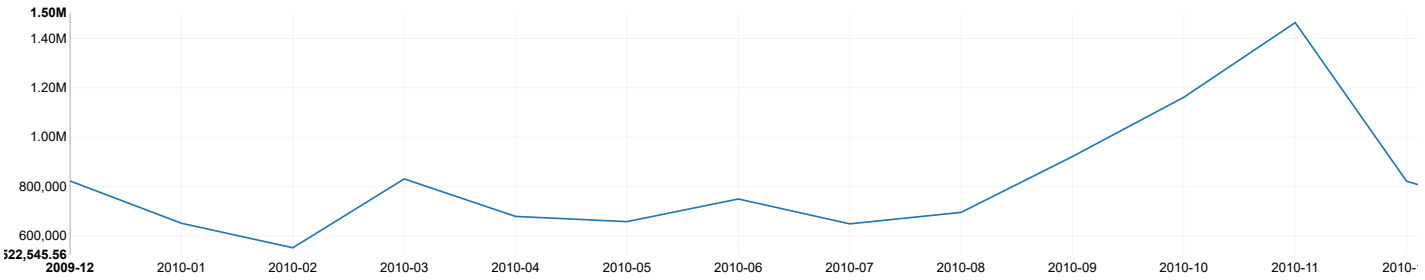
```
%pyspark
from pyspark.sql import functions as F

#Take clean data
df_clean = spark.table("online_retail_final")


##Feature Enginerring dalam zappelin gune pyspark
df_clean = df_clean.withColumn("Order_Month", F.date_format(F.col("InvoiceDate"), "yyyy-MM")) \
                   .withColumn("Order_Date", F.to_date(F.col("InvoiceDate"))) \
                   .withColumn("Order_Hour", F.hour(F.col("InvoiceDate"))) \
                   .withColumn("Day_Of_Week", F.dayofweek(F.col("InvoiceDate"))) \
                   .withColumn("Day_Type", F.when(F.col("Day_Of_Week").isin(1, 7), "Weekend").otherwise("Weekday")) \
                   .withColumn("Time_Period",
                       F.when((F.col("Order_Hour") >= 5) & (F.col("Order_Hour") < 12), "Morning")
                        .when((F.col("Order_Hour") >= 12) & (F.col("Order_Hour") < 18), "Afternoon")
                        .when((F.col("Order_Hour") >= 18) & (F.col("Order_Hour") < 23), "Evening")
                        .otherwise("Late Night")
                   )

##put as %sql table
df_clean.createOrReplaceTempView("online_retail_final_metrics")
```

READY

```
%sql
-- Compute Gross sales by Month
SELECT Order_Month, SUM(Price * Quantity) AS Total_Sale
FROM online_retail_final_metrics
GROUP BY Order_Month
ORDER BY Order_Month
```

READY

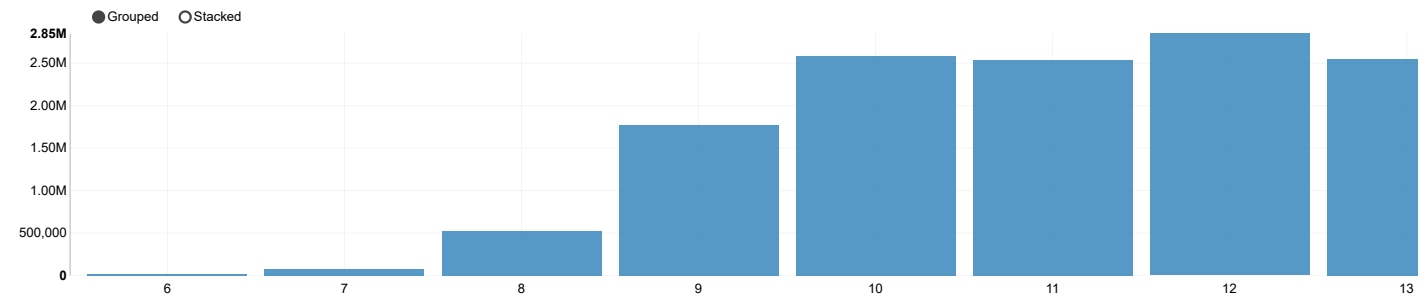
settings




```
%sql
-- Compute Gross sales by Hour

SELECT Order_Hour, SUM(Price * Quantity) AS Total_Sale
FROM online_retail_final_metrics
GROUP BY Order_Hour
ORDER BY Order_Hour
```

READY

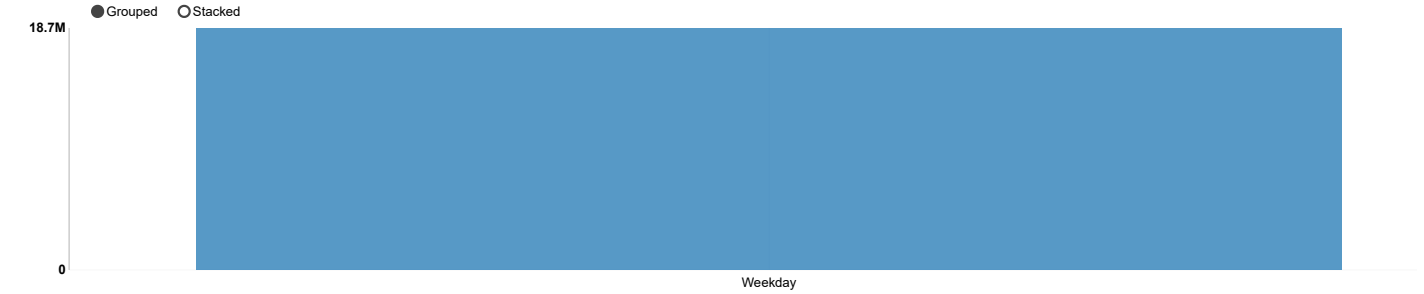
settings




```
%sql
-- Compute Gross sales by Week of Day
SELECT Day_Type, SUM(Price * Quantity) AS Total_Sales, COUNT(Invoice) AS Total_Orders
FROM online_retail_final_metrics
```

READY

settings




```
%sql
-- Compute Gross sales by Breakdown of time

SELECT Time_Period, SUM(Price * Quantity) AS Total_Sales, COUNT(Invoice) AS Total_Orders
FROM online_retail_final_metrics
GROUP BY Time_Period
ORDER BY Total_Sales DESC
```

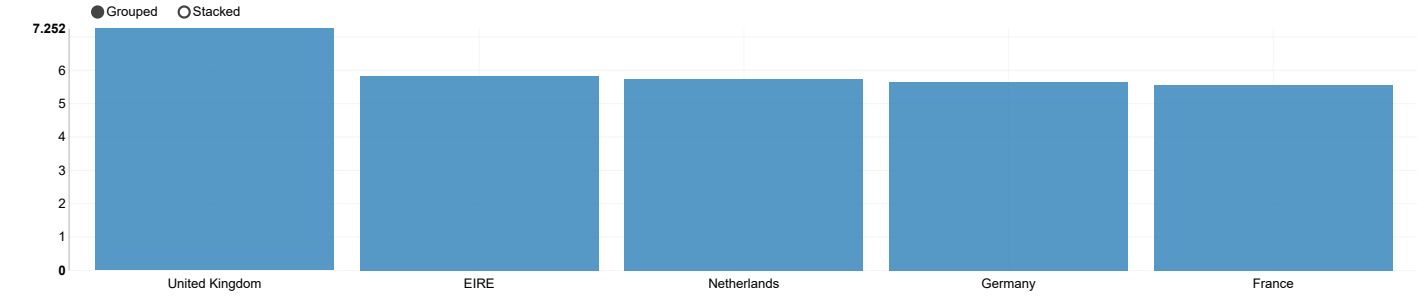
READY

settings


```
%sql
-- Compute Gross sales by Country (Top 10 Country)

SELECT Country, LOG10(SUM(Total_Sales)) as Log_Total_Sales
FROM online_retail
GROUP BY Country
ORDER BY Log_Total_Sales DESC
LIMIT 10
```

READY

settings




```
%sql
-- Compute Purchase Trend between country
WITH country_item_sales AS (
    SELECT
        Country,
        StockCode,
        MAX(Description) AS Product_Description,
        SUM(Quantity) AS Total_Quantity_Sold,
        ROW_NUMBER() OVER (PARTITION BY Country ORDER BY SUM(Quantity) DESC) as rank
    FROM online_retail_final_metrics
    -- Kita tapis keluar kuantiti negatif dan item NA jika ada
    WHERE Quantity > 0 AND Description != 'NA'
    GROUP BY Country, StockCode
)
SELECT
    Country,
    rank,
    StockCode,
    Product_Description,
    Total_Quantity_Sold
FROM country_item_sales
WHERE rank <= 10
ORDER BY Country, rank
```

READY

| Country | rank | StockCode | Product_Description |
|---|---|---|---|
| Australia | 1 | 22492 | MINI PAINT SET VINTAGE BOY+ GIRL |
| Australia | 2 | 23084 | RABBIT NIGHT LIGHT |
| Australia | 3 | 21915 | RED HARMONICA IN BOX |
| Australia | 4 | 21731 | RED TOADSTOOL LED NIGHT LIGHT |
| Australia | 5 | 22969 | HOMEMADE JAM SCENTED CANDLES |
| Australia | 6 | 22630 | DOLLY GIRL LUNCH BOX |
| Australia | 7 | 22629 | SPACEBOY LUNCH BOX |
| Australia | 8 | 22544 | MINI JIGSAW SPACEBOY |
| Australia | 9 | 22539 | MINI JIGSAW DOLLY GIRL |

READY

```
%sql
-- Compute Purchase Trend Similarities between country


WITH country_top_10 AS (
    SELECT
        Country,
        StockCode,
        MAX(Description) AS Product_Description,
        SUM(Quantity) AS Total_Quantity_Sold,
        ROW_NUMBER() OVER (PARTITION BY Country ORDER BY SUM(Quantity) DESC) as rank
    FROM online_retail_final_metrics
    WHERE Quantity > 0 AND Description != 'NA'
    GROUP BY Country, StockCode
)
SELECT
    StockCode,
    MAX(Product_Description) AS Product_Name,
    COUNT(DISTINCT Country) AS Number_Of_Countries_As_Top_10,
    COLLECT_SET(Country) AS List_Of_Countries
FROM country_top_10
WHERE rank <= 10
GROUP BY StockCode
HAVING COUNT(DISTINCT Country) > 1
ORDER BY Number_Of_Countries_As_Top_10 DESC
```

| StockCode | Product_Name | Number_Of_Countries_As_Top_10 | List_Of_Countries |
|---|---|---|---|
| 21240 | [illegible] | 2 | WrappedArray(RSA, Hong Kong) |
| 22423 | REGENCY CAKESTAND 3 TIER | 2 | WrappedArray(Brazil, Lebanon) |
| 21915 | RED HARMONICA IN BOX | 2 | WrappedArray(Australia, Lithuania) |
| 17003 | BROCADE RING PURSE | 2 | WrappedArray(West Indies, United Kingdom) |
| 22307 | GOLD MUG BONE CHINA TREE OF LIFE | 2 | WrappedArray(Malta, Lithuania) |
| 22582 | PACK OF 6 SWEETIE GIFT BOXES | 2 | WrappedArray(Austria, Nigeria) |
| 20677 | PINK SPOTTY BOWL | 2 | WrappedArray(RSA, Hong Kong) |
| 21928 | JUMBO BAG SCANDINAVIAN PAISLEY | 2 | WrappedArray(Portugal, Channel Islands) |
| 23077 | DOUGHNUT LIP GLOSS | 2 | WrappedArray(Bahrain, Finland) |

READY

```
%sql
-- Compute top spender (High Value) customer
SELECT
    Customer_ID,
    Country,
    COUNT(DISTINCT Invoice) AS Total_Orders,
    SUM(Quantity) AS Total_Items_Bought,
    ROUND(SUM(Price * Quantity), 2) AS Total_Amount_Spent
FROM online_retail_final_metrics
WHERE Customer_ID IS NOT NULL
  AND Price > 0
  AND Quantity > 0
GROUP BY Customer_ID, Country
ORDER BY Total_Amount_Spent DESC
LIMIT 10
```

settings

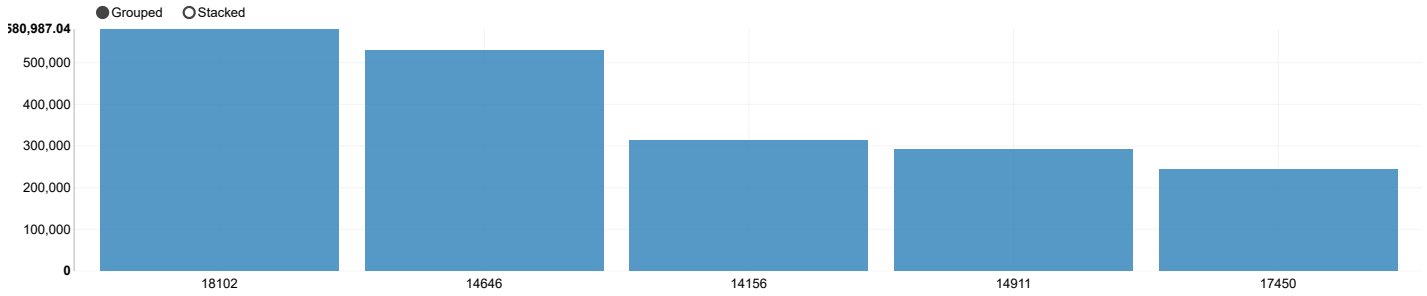
All fields:

Customer_ID | Country | Total_Orders | Total_Items_Bought | Total_Amount_Spent

Keys: Customer_ID

Groups:

Values: Total_Amount_Spent SUM




```
%sql
-- Compute AOV by Country

WITH invoice_totals AS (
    SELECT
        Country,
        Invoice,
        SUM(Price * Quantity) AS Invoice_Total_Value
    FROM online_retail_final_metrics
    WHERE Price > 0 AND Quantity > 0
    GROUP BY Country, Invoice
)
SELECT
    Country,
    COUNT(Invoice) AS Total_Invoices,
    ROUND(AVG(Invoice_Total_Value), 2) AS Average_Order_Value,
    ROUND(SUM(Invoice_Total_Value), 2) AS Total_Country_Revenue
FROM invoice_totals
GROUP BY Country
HAVING Total_Invoices > 10
ORDER BY Average_Order_Value DESC
```

READY

| Country | Total_Invoices | Average_Order_Value |
|---|---|---|
| Netherlands | 228 | 2429.99 |
| Singapore | 11 | 2301.55 |
| Australia | 95 | 1781.93 |
| Denmark | 43 | 1594.9 |
| Hong Kong | 15 | 1579.03 |
| Japan | 33 | 1303.75 |
| Norway | 45 | 1251.61 |
| Switzerland | 93 | 1082.64 |
| Greece | 18 | 1060.9 |

```
%sql
-- Top Country is UK but Top AOV is Netherland
-- Check if really UK is mostly B2C and Netherland B2B

WITH invoice_summary AS (
    SELECT
        Country,
        Invoice,
        SUM(Quantity) AS Total_Items_In_Basket,
        SUM(Price * Quantity) AS Invoice_Value
    FROM online_retail_final_metrics
    WHERE Price > 0 AND Quantity > 0
    GROUP BY Country, Invoice
)
SELECT
    Country,
    COUNT(Invoice) AS Total_Invoices,
    ROUND(AVG(Total_Items_In_Basket), 0) AS Avg_Items_Per_Invoice,
    ROUND(AVG(Invoice_Value), 2) AS Average_Order_Value
FROM invoice_summary
WHERE Country IN ('United Kingdom', 'Netherlands')
GROUP BY Country
```

READY

| Country | Total_Invoices | Avg_Items_Per_Invoice |
|---|---|---|
| United Kingdom | 36536 | 251.0 |
| Netherlands | 228 | 1684.0 |

```
%sql
-- Top purchase item bulk purchase in Netherland
SELECT
```

READY

```
    StockCode,
    MAX(Description) AS Product_Description,
    SUM(Quantity) AS Total_Quantity_Sold,
    COUNT(DISTINCT Invoice) AS How_Many_Invoices,
    ROUND(SUM(Price * Quantity), 2) AS Total_Revenue
FROM online_retail_final_metrics
WHERE Country = 'Netherlands'
  AND Quantity > 0
  AND Description != 'NA'
GROUP BY StockCode
ORDER BY Total_Quantity_Sold DESC
```

settings

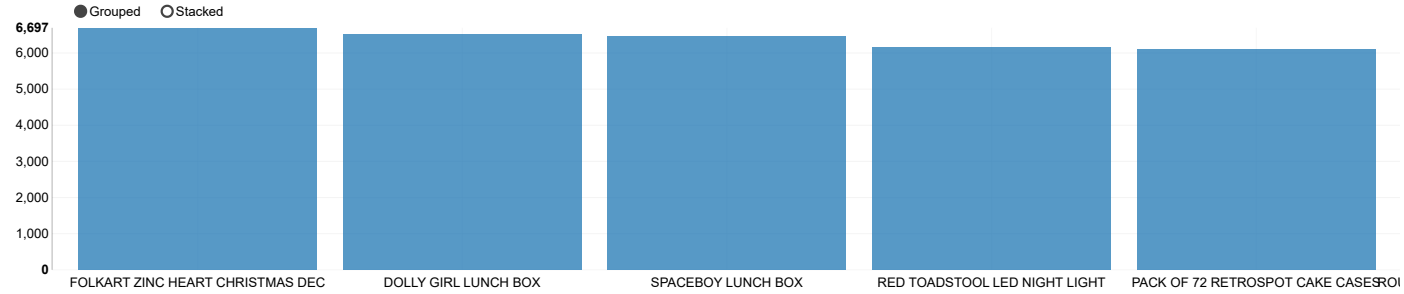
All fields:

StockCode | Product_Description | Total_Quantity_Sold | How_Many_Invoices | Total_Revenue

Keys: Product_Description

Groups:

Values: Total_Quantity_Sold SUM




```
%sql
```

READY